SIH 2026 PS 26017 — Predictive Analytics Model for Early Detection of Land Acquisition Delays

Research, data-source, mathematical-model and Claude handoff pack

Executive conclusion

This pack gives a defensible route for building the SIH 26017 prototype. The strongest foundation is the combination of India's MoSPI/PAIMANA project-monitoring data with the published South Asia infrastructure-delay research by Andri■ et al. The 2025 delay paper uses 138 completed projects, including 83 Indian projects, and compares linear, quadratic and Random Forest regression. The related 2024 paper states that 108 of the 138 projects came from ADB completion reports and 30 from MoSPI.

Important: the published papers establish the existence and provenance of the 138-project database, but this research did not locate an official downloadable 138-row CSV. Therefore, the team should not claim to possess that exact dataset unless it is obtained from the authors/supplementary files. The safe approach is to reconstruct records from the cited ADB/MoSPI source reports and separately label any synthetic land-acquisition fields.

1. Primary real-data sources

Source	What it provides	Use
MoSPI PAIMANA	Central-sector project monitoring; project counts, costs, schedules, progress and delay/slippage information	Primary Indian government source
MoSPI Public Dashboard	Project-monitoring dashboard with sector/state/project-cost filters and aggregate/project information	Exploration + validation
MoSPI Flash Reports	Monthly reports; project monitoring and reported reasons/issues for slippage	Extract project-level historical observations
ADB completion reports	Completed South Asian infrastructure projects with project name, country, type, estimated/actual cost, planned duration, implementation p	Reconstruct the 138-project research dataset
Devi & Sindhu 2025	Indian infrastructure delay factors; 72-professional survey; RII/PCA	Feature justification and optional survey layer

2. Evidence from the key papers

Andri■ et al. 2025: 138 completed South Asian projects; 83 are in India. The data includes project name, country, decision year, project type, estimated cost, actual cost, planned duration, implementation period and causes of delays. The study reports an average time overrun of 86.66% and identifies land acquisition/resettlement, procurement delay, contractor delay and design revision among major root causes. Random Forest regression is studied as a flexible non-parametric model.

Andri■ et al. 2024: the same 138-project database is described in greater detail: 108 projects were taken from ADB and 30 from MoSPI. The paper studies cost overruns and reports Random Forest as the most suitable of the tested functions for the relevant cost-overrun relationships.

Devi & Sindhu 2025: survey of 72 construction professionals; RII was used to rank 25 delay causes. Land acquisition was especially important for road projects (RII=0.68), and PCA grouped causes into Resource & Supply Chain, Government & External, Contractor Management, and Design & Planning. The paper reports VIF values below 5 and a four-factor solution explaining 68.3% of variance.

3. Mathematical formulas to implement

- Time-overrun percentage = 100 × (actual implementation period – planned duration) / planned duration
- Cost-overrun percentage = 100 × (actual cost – estimated/original cost) / estimated/original cost
- Acquisition completion (%) = 100 × land acquired / land required
- Compensation paid (%) = 100 × compensation paid / compensation required
- Linear regression: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \varepsilon$
- Logistic regression: $P(\text{delay})=1/(1+e^{-z})$, where $z=\beta_0+\Sigma\beta_i x_i$
- RII = $\Sigma W / (A \times N)$, with A=5 for a 5-point Likert scale
- Optional normalized legal-dispute intensity = pending legal cases / (affected parcels + 1)

4. Recommended dataset

Use the attached CSV schema as the master data dictionary. The most important distinction is provenance:

REAL	Directly observed in ADB/MoSPI/project documents; keep source URL/document.
DERIVED	Computed from real fields, e.g. acquisition %, time-overrun %, cost-overrun %.
SYNTHETIC	Only for prototype/demo where no public project-level land-acquisition field is available; never represent it as government data.

Core model features: project type, state, estimated cost, planned duration, implementation period, land-acquisition delay indicator, acquisition %, compensation paid %, affected parcels/families, pending legal cases, R&R; completion %, clearance issues, encroachment/ROW issues, approval delay and historical delay statistics.

5. ML plan for SIH

Stage	Method	Purpose
Baseline 1	Linear regression	Predict time-overrun months or percentage; establish an interpretable baseline.
Baseline 2	Logistic regression	Predict delayed/not delayed and probability of delay.
Main	Random Forest / XGBoost	Capture nonlinear relationships and interactions.
Explainability	SHAP / permutation importance	Show why a project received a high-risk prediction.
Validation	Cross-validation + held-out test set	MAE/RMSE/R^2 for regression; precision/recall/F1/ROC-AUC for classification.
Leakage control	Remove post-outcome variables	Do not feed actual completion information into a model intended for early prediction.

6. Recommended SIH output

The dashboard should not only say 'delayed'. It should output: (1) delay probability, (2) predicted months of delay, (3) risk band, (4) top contributing factors, (5) affected acquisition stage, and (6) recommended intervention. Example: 84% predicted risk; 9.8 months predicted delay; top drivers = low acquisition completion, pending legal cases and low compensation completion. This is a product design recommendation, not a published numerical rule.

7. Data acquisition workflow

1. Download/collect MoSPI Flash Reports and public dashboard observations.

2. Identify completed/ongoing projects relevant to roads, highways, railways and other infrastructure.

3. Reconstruct the 138-project research dataset from ADB/MoSPI source documents if the authors' dataset is unavailable.

4. Normalize project names, dates, costs, durations and sectors.

5. Extract delay-cause text and create binary indicators such as land acquisition, resettlement, procurement, contractor and design revision.

6. Where project-level land-acquisition metrics are available, add acquisition %, compensation %, legal cases, R&R; %, etc.

7. Create derived variables and preserve source provenance.

8. Train baseline and main models; perform leakage checks and cross-validation.

9. Use SHAP/feature importance to generate human-readable explanations.

10. Only after real-data coverage is assessed, add a clearly labelled synthetic demo layer for missing land-acquisition fields.

8. Links / references

MoSPI PAIMANA: https://paimana-proj.mospi.gov.in/

MoSPI Public Dashboard: https://ipm.mospi.gov.in/Home/PublicDashboard

MoSPI IPMD Project Monitoring: https://ipm.mospi.gov.in/EditOcms/DynamicPages?pageid=34

MoSPI FAQ/OCMS: https://ipm.mospi.gov.in/FAQ/Index

MoSPI report archive: https://ipm.mospi.gov.in/WhatsNewViewMore/ViewMore

Andri■ et al. 2025 — time delays: https://www.sciencedirect.com/science/article/pii/S1226798825003241

Andri■ et al. 2024 — cost overruns: https://www.mdpi.com/2071-1050/16/24/11159

Devi & Sindhu 2025 — India delays: https://link.springer.com/article/10.1007/s40030-025-00899-5

Devi & Sindhu institutional record: https://impressions.manipal.edu/open-access-archive/12698

9. Ready-to-send Claude instructions

A full handoff prompt is included as CLAUDE_HANDOFF_PROMPT.txt in the accompanying pack. It tells Claude what sources to use, what data provenance rules to follow, which formulas to implement, and what model/dashboard outputs to build.

10. Source verification note

This pack was prepared from publicly accessible sources checked on 26 August 2026. MoSPI's current PAIMANA pages state that IPMD monitors projects costing ■150 crore and above and captures progress, financial parameters and reasons/issues for schedule and cost slippages. The public dashboard displayed approximately 1,987 monitored projects in its May 2026 view. Current portal figures can change, so treat them as time-stamped observations rather than fixed constants.